Project Topic: (Agnese 1)

Nature-inspired patterns

Nature is a rich source of inspiration for the study and design of complex systems. Animals, even seemingly simple organisms, are capable of building remarkable structures and interacting with each other through complex collective behaviors. Such natural patterns have often inspired engineering solutions, computational methods, and models of social organization.

In this project, you will identify a nature-inspired complex pattern and develop a simulation to reproduce and investigate it. For example, how can the construction of a spider web be modeled? How does the collective behavior of an ant colony emerge from simple individual rules? The goal is to explore how relatively simple mechanisms and interactions can give rise to complex structures or behaviors, and to assess how effectively they can be captured through simulation.



Fig.1 Photo of a spider web. Spider webs are remarkable natural architectures for capturing prey. Vibrations propagating through the silk allow the spider to detect and locate captured prey, providing an elegant example of how complex functionality can emerge from structure and material properties. *Photo by Rafael Garcin on Unsplash*

References:

[1] Aoyanagi, Y. and Okumura, K. *Simple Model for the Mechanics of Spider Webs*, Phys. Rev. Lett. **104**, 038102 (2018).
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[2] Freas, C.A., Buehlmann, C. & Spetch, M.L. *Combining social and private information: How ants use pheromones and learnt cues to navigate*. Learn. Behav. **54**, 23–36 (2026).
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